

ASSESSMENT-1
CSA1707- ARTIFICIAL INTELLIGENCE

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MCQ Test

Q1. In informed search strategies, some algorithms use only the heuristic value to decide which node to expand next. This approach focuses on selecting the node that appears closest to the goal without considering the cost already incurred. Which algorithm follows this principle?

Answer:

Greedy Best-First Search (Option C)

Explanation:

Greedy Best-First Search selects the node with the lowest heuristic value $h(n)$. It considers only the estimated distance to the goal and ignores the path cost already travelled, making it faster but not always optimal.

Q2. A search is widely used in pathfinding and graph traversal because it combines the actual cost from the start node and the estimated cost to the goal. What is the evaluation function used in A search?*

Answer:

$f(n) = g(n) + h(n)$ (Option C)

Explanation:

In A* Search:

- **$g(n)$** = Actual cost from the start node.
- **$h(n)$** = Estimated cost to the goal.
- **$f(n)$** = Total estimated cost.

This evaluation function helps A* find the optimal path efficiently.

Q3. Heuristic functions play a crucial role in informed search by guiding the algorithm toward the goal. What property ensures optimality in A search?*

Answer:

Admissibility (Option B)

Explanation:

A heuristic is **admissible** if it never overestimates the actual cost to reach the goal. An admissible heuristic guarantees that A* Search always finds the optimal solution.

Q4. Local search algorithms operate by starting with an initial solution and iteratively improving it based on a given objective function. Which of the following is a local search algorithm?

Answer:

Hill Climbing (Option B)

Explanation:

Hill Climbing is a local search algorithm that continuously moves toward a better neighboring state until no better solution is available. It is commonly used for optimization problems.

Q5. In adversarial search, agents compete against each other in a game environment. Which algorithm is commonly used for such decision-making?

Answer:

Minimax Algorithm (Option B)

Explanation:

The Minimax Algorithm is used in two-player games such as chess and tic-tac-toe. It evaluates possible moves by considering both the player's and the opponent's actions to select the optimal move.

Q6. A delivery robot in a warehouse selects paths based only on the estimated distance to the destination without considering obstacles already encountered. Which algorithm is the robot using?

Answer:

Greedy Best-First Search (Option B)

Explanation:

Greedy Best-First Search selects the next node based only on the heuristic value $h(n)$, which estimates the distance to the goal. Since it ignores the actual path cost, it may produce faster but non-optimal solutions.

Q7. A GPS navigation system calculates routes by combining the distance already travelled and an estimate of the remaining distance to the destination. Which algorithm best represents this approach?

Answer:

*A Search (Option B)**

Explanation:

A* Search combines the actual path cost $g(n)$ and the heuristic estimate $h(n)$ to calculate $f(n) = g(n) + h(n)$. This allows it to efficiently find the shortest and optimal path.

Q8. A university is scheduling exams for multiple subjects with constraints such as no overlapping exams and limited rooms. This problem is best modeled as:

Answer:

Constraint Satisfaction Problem (Option B)

Explanation:

A Constraint Satisfaction Problem (CSP) consists of variables, domains, and constraints. Exam scheduling is a CSP because all constraints, such as room availability and avoiding exam clashes, must be satisfied.

Q9. An AI-based chess program evaluates all possible moves and anticipates the opponent's responses before making a decision. Which algorithm is being applied here?

Answer:

Minimax Algorithm (Option C)

Explanation:

The Minimax Algorithm is used in two-player games. It evaluates all possible moves by considering both the maximizing player's and minimizing opponent's actions to choose the best move.

Q10. A robot explores an unknown environment where it does not have prior knowledge of obstacles. It makes decisions step-by-step and updates its knowledge as it moves. Which type of agent is suitable for this scenario?

Answer:

Online Search Agent (Option B)

Explanation:

An Online Search Agent explores the environment while simultaneously gathering information. It updates its knowledge after each move, making it suitable for unknown or partially observable environments.

Q11. A search guarantees optimal solutions only when certain conditions are satisfied by the heuristic function. Why is an admissible heuristic important?*

Answer:

It ensures optimality (Option B)

Explanation:

An admissible heuristic never overestimates the actual cost to reach the goal. This property guarantees that the A* Search algorithm always finds the optimal path. Therefore, admissibility is essential for ensuring optimal solutions.

Q12. In large game trees, evaluating every node becomes computationally expensive. What is the primary benefit of Alpha-Beta Pruning?

Answer:

Reduces number of nodes evaluated (Option B)

Explanation:

Alpha-Beta Pruning eliminates branches that cannot influence the final decision in the Minimax algorithm. This reduces the number of nodes evaluated, making game-tree search faster while producing the same optimal result.

Q13. Hill Climbing algorithm continuously moves toward better states based on local improvements. What is the main drawback of Hill Climbing?

Answer:

Getting stuck in local maxima (Option C)

Explanation:

Hill Climbing always moves toward a better neighboring state. However, it may stop at a local maximum where no neighboring state is better, even though a better global solution exists.

Q14. Constraint Satisfaction Problems involve variables, domains, and constraints. What ensures a valid CSP solution?

Answer:

Constraint satisfaction (Option C)

Explanation:

A Constraint Satisfaction Problem (CSP) is solved only when all variables are assigned values without violating any constraints. Therefore, satisfying all constraints ensures a valid solution.

Q15. In adversarial games, evaluation functions are used when it is not feasible to explore the entire game tree. What is the role of an evaluation function?

Answer:

Estimate utility of a state (Option B)

Explanation:

An evaluation function estimates the desirability or utility of a game state. It helps algorithms like Minimax choose the best move when searching the complete game tree is not practical.

Q16. In A search, the evaluation function is calculated using both the path cost and heuristic value. Suppose a node has a path cost of 7 and a heuristic estimate of 5. What is the value of $f(n)$?*

Answer:

12 (Option A)

Explanation:

The A* evaluation function is:

$$f(n) = g(n) + h(n)$$

where:

- $g(n) = 7$
- $h(n) = 5$

Therefore,

$$f(n) = 7 + 5 = 12$$

Q17. A heuristic function sometimes overestimates the actual cost to reach the goal. What happens in such a case?

Answer:

*A becomes non-optimal (Option B)**

Explanation:

If the heuristic overestimates the actual cost, it is no longer admissible. As a result, A* Search may still find a solution, but it is not guaranteed to be the optimal one.

Q18. Consider a search tree with a branching factor of 2 and depth of 4. What is the approximate number of nodes?

Answer:

31 (Option C)

Explanation:

For a full binary tree,

Total Nodes = $2^{(4+1)} - 1$

= $2^5 - 1$

= $32 - 1$

= **31**

Hence, the approximate number of nodes is **31**.

Q19. In a minimax tree, each node has 3 children and the depth of the tree is 3. How many leaf nodes are present?

Answer:

27 (Option B)

Explanation:

The number of leaf nodes is calculated using:

Leaf Nodes = **Branching Factor**^{Depth}

= 3^3

= **27**

Therefore, the minimax tree contains **27 leaf nodes**.

Q20. Alpha-Beta Pruning improves the efficiency of Minimax by reducing the number of nodes evaluated. What is the best-case time complexity?

Answer:

$O(b^{d/2})$ (Option B)

Explanation:

In the best case, Alpha-Beta Pruning eliminates many unnecessary branches, reducing the effective search space. Therefore, its best-case time complexity is **$O(b^{d/2})$** , making it much more efficient than the standard Minimax algorithm.